

ESSENTIALS

Mathematics	Linear algebra, Calculus, ODE/PDE, Probability theory, Information theory, Numerical analysis, Optimization, Multivariate regression, PDF, KDE methods, Dimensionality reduction, Loss functions, Metrics (precision, recall, F1 score, AUC-ROC)
Modeling methods	Regression, Random Forests, kNN, SVM, PCA, Time Series Models, Neural Networks (CNN, RNN, LSTM, PINNS), VAE, XGBoost, Time series, Complex Networks
Tools	Python, R studio, SQL, PostgreSQL, MySQL, MongoDB, Azure, Git, C++, LLM
Frameworks / Platforms	LangChain, FastAPI, Docker, Keras, Streamlit, Shiny, HPC, Angular, Databricks, Hugging Face
Visualization	Matplotlib, Seaborn, ggplot, Plotly, Superset, GnuPlot, Mathematica, MATLAB plots, NetworkX
Operating Systems	Linux, Windows

PROJETS

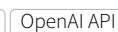
AI AGENT ASSISTANT

Developing smart agent systems capable of answering questions based on content, extracted from user-provided documents such as PDF, TXT, or Word files. This system leverages RAG and a vector database to deliver context-aware responses. Full-stack development including frontend and backend logics.

VOICE AI ASSISTANT

Built a real-time voice assistant capable of handling inbound calls and responding to user queries using natural language. Leveraged LLMs in combination with LiveKit Cloud and Telnyx.

VARIATIONAL AUTOENCODERS FOR CHAOTIC SYSTEMS

Implemented a Variational Autoencoder (VAE) to learn and generate trajectories of chaotic dynamical Henon-Heiles Hamiltonian system. The model captures latent structure from test/train data (generated by molecular dynamics) and reconstructs realistic patterns based on feature space. Emphasis is on improvement through regularization and validation techniques.

BIG DATA ANALYTICS

Transforming Big-data analytics scripts from Scala to PySpark

PROFESSIONAL EXPERIENCE

Data Engineer, IRAN, **Currently working (remote)**

- > ETL pipelines and stream data analysis
- > ML models.
- > Fraud detection

Machine Learning Engineer, UAE,

- > Implementing backend algorithms for matchmaking and intelligent chatbot agents using Python, FastAPI and LLMs.
- > Conducted business-level research and data analysis to extract actionable insights.

Data science Internship, LIS DATA CENTER, Luxembourg

- > Building interactive dashboards using Shiny for real-time monitoring.
- > Analyzed patterns and anomalies in Time-Series data.
- > Consolidated multiple data sources into a centralized relational database.

R-Shiny MySQL Time-Series Dashboard Design

Support Researcher in Big Data Group, CS Departmen, UNIVERSITY OF LUXEMBOURG,

- > Managed Big data workflows on AWS EMR for processing large-scale datasets.
- > Migrated scripts from Scala to PySpark.

PySpark AWS EMR Scala Jupyter Big Data Distributed Computing

Research Assistant, DATA ANALYSIS AND MATHEMATICAL MODELLING, University of Luxembourg

- > Conducted research on Raman spectroscopy and light-matter interaction in quantum materials.
- > Modeled vibrational properties in 2D materials and diamond lattices.
- > Studied the behavior of superconducting materials under doping using density functional theory.
- > Conducting parallel calculations on HPC supercomputers.

Raman Spectroscopy DFT Quantum Mechanics Research Writing HPC

Teaching Assistant (TA), SHARIF UNIVERSITY OF TECHNOLOGY AND UNIVERSITY OF LUXEMBOURG,

- > Assisted in delivering undergraduate courses in Electricity and Quantum Mechanics.
- > Guided students through Mathematical physics courses. Methods II.

Physics Quantum Mechanics Teaching Communication Academic Support Mathematical Methods

EDUCATION

2022–2024 M.Sc. in Data Science, Université du Luxembourg, Luxembourg

2013–2015 M.Sc. in Physics, Sharif University of Technology, Tehran, Iran

2009–2013 B.Sc. in Physics, Semnan University, Semnan, Iran

COURSES

- > [Statistical Mechanics](#) (Ensembles, Partition function, entropy, Phase transitions, Ising model, ideal gas)
- > [Advanced Machine Learning](#) (Random forests, kernel methods, SVM, Neural networks, model evaluation, AUC-ROC, PCA)
- > [Non-Parametric Statistics](#) (KDE, Nadaraya-Watson estimator, non-parametric regression, Leave one out)
- > [High-Dimensional Statistics](#) (LASSO, Ridge regression, cross-validation, PCA, covariance matrix)
- > [Electrodynamics](#) (Maxwell's equations, vector and scalar potentials, gauge transformations, Green's functions)
- > [Complex Networks](#) (adjacency matrices, degree distribution, scale-free networks, betweenness, epidemic modeling)
- > [Natural Language Processing](#) (tokenization, stemming, lemmatization, POS, NER, TF-IDF, word embeddings)
- > [Deep Learning](#) (feedforward networks, backpropagation, activation functions, loss functions, SGD)
- > [Mathematical Physics](#) (differential equations, Fourier series, Legendre, Bessel, Hermite, tensor, complex analysis)
- > [Optimization and Probability](#) (convex optimization, Lagrange multipliers, KKT conditions, Markov chains)
- > [Time-Series Analysis](#) (stationarity, autocorrelation, ACF/PACF plots, ARIMA models, seasonal decomposition)
- > [Numerical Physics](#) (finite element methods, solving ODEs, numerical integration, Runge-Kutta methods)

LANGUAGES

German ● ● ● ○ ○
English ● ● ● ● ●
Persian ● ● ● ● ●

STRENGTHS AND SOFT SKILLS

- > Dedicated
- > Good communicator
- > Problem solver
- > Self-motivated and autonomous